Read the symbols without holding them in memory

X (random variable); x_i (one observation); $\bar{x}$ (sample mean); μ (population mean); s (sample SD); σ (population SD); s^2, σ^2 (variances, squared units); n (sample size); m (groups, except when explicitly event count); ν (df). A bar means average; a hat means estimate; $\sum$ means add. **Variance is not SD. Square SD for a variance; square-root variance for SD.**

Population (all units measured):

$$\mu = \frac{\sum_{i=1}^{N} x_i}{N}, \quad \sigma^2 = \frac{\sum_{i=1}^{N} (x_i - \mu)^2}{N}$$

Choose the outcome scale first

Interval-continuous: height, LDL, blood pressure. **Interval-discrete:** tender-joint count; age recorded in whole years. **Nominal:** ZIP code, country of birth (labels; no amount/order). **Ordinal:** cancer stage or pain category (ordered; gaps need not be equal). **No arithmetic meaning just because digits appear.**

Raw-data summary: freshman heights

Given $65, 69, 59, 64, 62, 67, 68, 62, 64, 65$ inches; $n = 10$. **1. Mean:**

$$\bar{x} = \frac{645}{10} = 64.5$$

2. Subtract the mean, then square each distance:

x_i	65	69	59	64	62
$x_i - \bar{x}$	.5	4.5	−5.5	−.5	−2.5
$(x_i - \bar{x})^2$	.25	20.25	30.25	.25	6.25

x_i	67	68	62	64	65
$x_i - \bar{x}$	2.5	3.5	−2.5	−.5	.5
$(x_i - \bar{x})^2$	6.25	12.25	6.25	.25	.25

3. Add the squared column; 4. divide; 5. square-root:

$$\sum (x_i - \bar{x})^2 = 82.5$$

$$s^2 = \frac{82.5}{9} = 9.1667, \quad s = \sqrt{9.1667} = 3.0277$$

$$\text{SEM} = \frac{3.0277}{\sqrt{10}} = 0.95743 \text{ in.}$$

SD (individual spread) is 3.0277 inches; SEM (uncertainty in the mean) is 0.95743 inches. **Whole population:** use N and μ, dividing squared distances by N. **Quick arithmetic check:**

$$\sum (x_i - \bar{x})^2 = \sum x_i^2 - \frac{(\sum x_i)^2}{n}$$

$$41685 - \frac{645^2}{10} = 82.5$$

Median, quartiles, interpolation

Sort first. The senior sample sorts to $66, 67, 69, 69, 70, 70, 71, 71, 72, 72$. For the 75th percentile:

$$R = \frac{75(10+1)}{100} = 8.25$$

$$x_{.75} = x_{(8)} + .25(x_{(9)} - x_{(8)})$$

$$= 71 + .25(72 - 71) = 71.25 \text{ in.}$$

Rank 8.25 is a location; 71.25 inches is the answer.

$$Q_1 = 68.5, \quad \text{IQR} = 71.25 - 68.5 = 2.75$$

Range endpoints are $[66, 72]$; numerical width is 6. Q–Q: approximately straight means Normal shape is plausible; systematic curvature suggests a mismatch.

$$\text{median} = \begin{cases} x_{((n+1)/2)} & n \text{ odd} \\ \dfrac{x_{(n/2)} + x_{(n/2+1)}}{2} & n \text{ even} \end{cases}$$

Discrete: validity → CDF → moments

Checks: every probability is nonnegative, and the total is one. Candidate $.4, .1, .4$ example totals $.9$, so it fails. **Probability table:**

x	−1	2	3	7
$f(x)$	.55	.15	.10	.20
$xf(x)$	−.55	.30	.30	1.40
$x^2 f(x)$	.55	.60	.90	9.80

$$\sum f(x) = 1, \quad E[X] = -.55 + .30 + .30 + 1.40 = 1.45$$

$$E[X^2] = .55 + .60 + .90 + 9.80 = 11.85$$

$$\text{Var}(X) = 11.85 - (1.45)^2 = 9.7475$$

$E[X^2]$ (average square) differs from $(E[X])^2$ (squared average). Square values, not their probabilities. General transformed mean:

$$E[h(X)] = \sum_x h(x) f(x)$$

CDF: add masses through the cutoff; the graph is a staircase:

$$F(x) = \begin{cases} 0 & x < -1 \\ .55 & -1 \le x < 2 \\ .70 & 2 \le x < 3 \\ .80 & 3 \le x < 7 \\ 1 & x \ge 7 \end{cases}$$

For example, $F(2) = .70$, while $P(X < 2) = .55$. At a jump, equality matters. **Sketch:** isolated probability bars at supported values; CDF flat between jumps, filled at the new height.

If asked for expected squared distance

Use the same mean $\mu = 1.45$. Each distance is squared before it is weighted:

$$E[(X-\mu)^2] = .55(-1 - 1.45)^2 + .15(2 - 1.45)^2$$
$$+ .10(3 - 1.45)^2 + .20(7 - 1.45)^2$$
$$= .55(6.0025) + .15(.3025) + .10(2.4025) + .20(30.8025)$$
$$= 3.301375 + .045375 + .24025 + 6.1605 = 9.7475$$

Same answer as mean square minus squared mean. Use the shorter form unless the question requests the distance calculation.

Discrete CDF: include the correct endpoint

$F(a^-)$: mass below a; $f(a)$: mass exactly at a.

$$P(X > a) = 1 - F(a), \quad P(X \ge a) = 1 - F(a^-)$$

$$P(a < X \le b) = F(b) - F(a)$$

$$P(a \le X \le b) = F(b) - F(a^-)$$

Include 2 and 3, exclude 7:

$$P(2 \le X < 7) = .15 + .10 = .25$$

$$= F(7^-) - F(2^-) = .80 - .55$$

Count outcomes; update dependent probabilities

Three fair coin tosses have eight equally likely sequences. Exactly two heads occurs as HHT, HTH, THH:

$$P(X = 2) = \frac{3}{8}$$

Two kings without replacement: after the first king, only three kings remain among 51 cards.

$$P(\text{two kings}) = \frac{4}{52}\frac{3}{51} = 0.0045249$$

Independence leaves the second probability unchanged. Without replacement, update what remains. General sequence rule:

$$P(A \cap B) = P(A) P(B \mid A)$$

Continuous density: area, not height

$f(x)$ is density height, not point probability. It is nonnegative and integrates to one. Example: $f(x) = x/2$ for $0 \le x \le 2$. **1. Check total area:**

$$\int_0^2 \frac{x}{2}\, dx = \left[\frac{x^2}{4}\right]_0^2 = 1$$

2. Accumulate area to a cutoff:

$$F(x) = \begin{cases} 0 & x < 0 \\ x^2/4 & 0 \le x \le 2 \\ 1 & x > 2 \end{cases}$$

3. Use the CDF for a left or right event:

$$P(X < 1) = \frac{1^2}{4} = .25$$

$$P(X > 1.5) = 1 - \frac{1.5^2}{4} = .4375$$

4. Mean: value × density; integrate:

$$E[X] = \int_0^2 x \frac{x}{2}\, dx = \left[\frac{x^3}{6}\right]_0^2 = \frac{4}{3}$$

5. Mean square, then variance:

$$E[X^2] = \int_0^2 \frac{x^3}{2}\, dx = \left[\frac{x^4}{8}\right]_0^2 = 2$$

$$\text{Var}(X) = 2 - \left(\frac{4}{3}\right)^2 = \frac{2}{9}$$

6. Median: half the probability, not half the support:

$$F(m) = \frac{1}{2} \Rightarrow \frac{m^2}{4} = \frac{1}{2} \Rightarrow m = \sqrt{2}$$

Any exact point has zero probability. For intervals use $F(b) - F(a)$. **Sketch:** rising straight-line density; CDF rises quadratically, then stays at one.

Unknown endpoint: normalize before using it

$f(x) = 0.1 + x$ on $0 < x \le c$, zero elsewhere. **Find the supported area first:**

$$1 = \int_0^c (.1 + x)\, dx = .1c + \frac{c^2}{2}$$

$$c^2 + .2c - 2 = 0$$

$$c = \frac{-.2 + \sqrt{.04 + 8}}{2} \approx 1.317745$$

The negative root does not fit positive support. Inside support:

$$F(x) = .1x + \frac{x^2}{2}$$

Outside support use zero before the start and one after the end. **Power integration:**

$$\int x^k\, dx = \frac{x^{k+1}}{k+1} + C \quad (k \ne -1)$$

Evaluate upper antiderivative minus lower antiderivative. A constant density is flat: probability is requested width ÷ full width.

Uniform density: the short formulas

Endpoints: a (lower), b (upper); total area one.

$$f(x) = \frac{1}{b-a} \quad (a < x \le b)$$

$$F(x) = \frac{x-a}{b-a} \quad (a < x \le b)$$

$$E[X] = \frac{a+b}{2}, \quad \text{Var}(X) = \frac{(b-a)^2}{12}$$

Valid CDF

$$0 \le F(x) \le 1, \quad F(-\infty) = 0, \quad F(\infty) = 1$$

Never decreases. Discrete jumps include their new mass; continuous CDFs accumulate density area.

Normal: shade → standardize → table

$X \sim N(\mu, \sigma^2)$: mean μ, variance σ^2, SD σ.

$$z = \frac{x-\mu}{\sigma}, \qquad \Phi(z) = P(Z \le z)$$

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right]$$

Use SD (not variance) in the denominator. The supplied z table lists positive z and left-tail areas.

$$\Phi(-z) = 1 - \Phi(z)$$

$$P(X > b) = 1 - \Phi(z_b)$$

$$P(a < X < b) = \Phi(z_b) - \Phi(z_a)$$

z_a (standardized lower cutoff); z_b (standardized upper cutoff). Same Normal curve: mean = median; about 68%, 95%, 99.7% lies within one, two, three SDs. Density is bell-shaped; CDF rises smoothly from zero to one.

Glucose probabilities: all steps

$\mu = 95$, $\sigma = 8.7$ mg/dL. Find probabilities at cutoffs 85 and 115.

$$z_{85} = \frac{85-95}{8.7} = -1.149425 \approx -1.15$$

$$z_{115} = \frac{115-95}{8.7} = 2.298851 \approx 2.30$$

Read the supplied table (rounded z):

$$\Phi(1.15) = .8749, \quad \Phi(2.30) = .9893$$

$$P(X < 85) = 1 - .8749 = .1251$$

$$P(85 \le X < 115) = .9893 - .1251 = .8642$$

$$P(X \ge 115) = 1 - .9893 = .0107$$

$$.1251 + .8642 + .0107 = 1$$

Full-precision calculator checks: $.125190, .864053, .0107567$. Table rounding explains the small differences. $P(X = 95) = 0$ because this is continuous. The population variance is $8.7^2 = 75.69$.

Percentile: area → z → original units

Find the 97.5th percentile of the same glucose distribution. **1. Cumulative area; 2. z; 3. reverse standardization:**

$$p = .975, \quad z_p = 1.96$$

$$x_p = \mu + z_p\sigma = 95 + 1.96(8.7)$$

$$x_{.975} \approx 112.052 \text{ mg/dL}$$

For quartiles, $z_{.25} \approx -.67449$, $z_{.75} \approx .67449$: cutoffs 89.1319 and 100.8681 mg/dL. **Direction:** cutoff given → area; probability given → inverse cutoff. Above-median percentiles must exceed the mean.

Standard Uniform: work the moments

Standard Uniform support: $0 < x \le 1$. Then $f(x) = 1$ and inside the support:

$$F(x) = \int_0^x 1\,du = x$$

$$E[X] = \int_0^1 x\,dx = \left[x^2/2\right]_0^1 = 1/2$$

$$E[X^2] = \int_0^1 x^2\,dx = \left[x^3/3\right]_0^1 = 1/3$$

$$\text{Var}(X) = \frac{1}{3} - \left(\frac{1}{2}\right)^2 = \frac{1}{12}$$

Short route for any interval inside support: requested width divided by full width. CDF is zero before support and one after it.

Average of measurements: use SEM

X is one measurement; $\bar{X}$ is the mean of an entire sample. Repeated samples give different means. Their center stays at μ, but their spread shrinks:

$$E[\bar{X}] = \mu, \quad \text{Var}(\bar{X}) = \frac{\sigma^2}{n}$$

$$\text{SD}(\bar{X}) = \frac{\sigma}{\sqrt{n}}$$

If individuals are Normal, means are Normal for any n. Otherwise the course's CLT guideline uses sufficiently large independent samples ($n \ge 30$). **Impurity sample mean:** $\mu = 4$, $\sigma = 1.5$ grams, $n = 50$; find $P(3.5 < \bar{X} < 3.8)$.

$$\text{SEM} = \frac{1.5}{\sqrt{50}} = .212132$$

$$z_L = \frac{3.5-4}{.212132} = -2.35702$$

$$z_U = \frac{3.8-4}{.212132} = -.942809$$

$$P = \Phi(-.942809) - \Phi(-2.35702)$$

$$\approx .172889 - .00921106 = .163678$$

With z-table rounding: use −2.36 and −0.94; approximate result .1644. **Mean in the question → SEM in the denominator.**

What changes the center or uncertainty?

Adding a constant shifts the center but leaves spread unchanged. Multiplying stretches SD by the coefficient magnitude and variance by its square.

$$E[aX + bY + c] = aE[X] + bE[Y] + c$$

$$\text{Var}(aX + bY + c) = a^2\sigma_X^2 + b^2\sigma_Y^2$$

The variance sum requires independent X, Y. Subtracting an independent measurement still adds uncertainty; do not subtract its variance. **Sample-size shortcut:** multiply n by four to halve SEM. **Variance-estimate uncertainty (Normal setting):**

$$\widehat{\text{SE}}(s^2) = s^2\sqrt{\frac{2}{n-1}}$$

s (individual SD); $s/\sqrt{n}$ (SE of the mean); the expression above is SE of the variance. Unbiased means correct over repeated samples, not exact in one sample.

Why the mean's denominator changes

n independent observations, each with variance σ^2: add their variances, then scale by the square of the averaging factor.

$$\text{Var}\left(\sum X_i\right) = n\sigma^2$$

$$\text{Var}(\bar{X}) = \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n}$$

$$\text{SD}(\bar{X}) = \frac{\sigma}{\sqrt{n}}$$

Use s to estimate unknown population SD. ANOVA multiplies the variance of group means by group size to put it back on the individual-variance scale.

Normal: turn wording into the shaded area

"Above" uses the right tail; **"between"** subtracts left areas; **"top tail"** starts from a right-tail area. For a cutoff with upper-tail probability q:

$$\Phi(z) = 1 - q, \quad x = \mu + \sigma z$$

For the middle 95% of a Normal population, leave .025 in each tail. The two measurement cutoffs are:

$$\mu - 1.96\sigma, \qquad \mu + 1.96\sigma$$

These cutoffs bound individual measurements. **Density height can exceed one; probability area cannot.**

Expected function: transform values, not probabilities

$$E[h(X)] = \int h(x)f(x)\,dx$$

Endpoint density $f(x) = .1 + x$, support $(0, c]$, with the earlier $c \approx 1.317745$:

$$E[X] = \int_0^c (.1x + x^2)\,dx = .05c^2 + \frac{c^3}{3}$$

$$\approx .849556, \quad E[2X + 1] = 2E[X] + 1 \approx 2.69911$$

For the Normal glucose example, shifting every value by 26 shifts only the mean: $E[X - 26] = 95 - 26 = 69$; the SD remains 8.7.

Read THESE supplied statistical tables

z table: left-tail area for positive z. Locate the whole decimal z value in the repeated column pairs; use symmetry for negative z. Example $z = .05$ gives $.5199$. **t table:** row ν (df), column total two-tail α. Do not halve alpha again for this table. At $\nu = 4$, column $.05$ gives 2.776. For two-sided z, the infinity-df row gives 1.960. **F table:** numerator df across the top, denominator df down the side. Lightface is $.05$, boldface $.01$; select the matching line. **χ² table:** row df, upper-tail column alpha. At $.05$, df 1 gives 3.841, df 2 gives 5.991. **Missing df:** bracket using the neighboring available rows; a result beyond both bounds has an unambiguous decision. Use the provided conventions for any tighter case. **Pack separately:** original unannotated z, t, F, χ² tables; table pages do not count against the four notes sides.

TI-84: only the useful non-basic entries

Raw summaries: STAT → EDIT → enter data in L1. To clear old entries, highlight the L1 name, CLEAR, ENTER. STAT → CALC → 1-Var Stats; List=L1; frequency blank/1; Calculate. Read x̄, **Sx (sample SD)** and n. σx uses the list-size denominator; do not substitute it for sample SD.

Normal area: 2nd → VARS (DISTR) → normalcdf. For glucose: lower 85; upper 115; μ 95; σ 8.7; Paste/ENTER. For the mean impurity: lower 3.5; upper 3.8; μ 4; σ 1.5/√50. That final input is SEM.

Percentile: DISTR → invNorm: area .975; μ 95; σ 8.7; LEFT tail if offered. For a tail extending to infinity, use the calculator's practical bound ±1E99, with E entered using 2nd → comma (EE).

One-sample t check: STAT → TESTS → T-Test → Stats: μ0 72; x̄ 76.8; Sx 4.15; n 5; choose ≠; Calculate.

Pooled two-sample check: 2-SampTTest → Stats: x̄1 82.2, Sx1 2.49, n1 25; x̄2 86.1, Sx2 2.09, n2 25; ≠; **Pooled: Yes**; Calculate.

Adjusted positive t cutoff: DISTR → invT, left area 1−.05/(2k), df from the method. If invT is absent on that OS, use the t table. **Two-tail p after hand t:** DISTR → tcdf, lower=|t|, upper=1E99, df=method's df; multiply the resulting tail area by two.

Retain intermediate values: after pooled variance, STO→ A (A holds pooled variance). Build the SE using A; STO→ B (B holds SE). Then enter the mean difference divided by B. Keep A/B's meanings written beside the work.

$$A = 5.2841, \quad B = \sqrt{A(1/25 + 1/25)}$$

$$t = (82.2 - 86.1)/B$$

Use the entire denominator in parentheses. A calculator result does not replace the written null, setup, df, and conclusion. Standard two-proportion/χ² calculator routines may omit the required Yates correction; use the hand corrected recipe for those.

Negative entries: use the (-) key for a negative bound. Use the subtraction key only for subtracting. Select functions from their menu, then check the inputs before calculating.

Write an exam answer like this

1. Test + H₀. 2. Labeled inputs. 3. Substitute and calculate. 4. df + table cutoff. 5. Decision in context. Two-sided t uses $|t|$; F and χ^2 use their upper tail. Reject when the appropriate statistic exceeds its critical value, or when $p < \alpha$. **Assumptions for course mean tests:** interval outcome, independent random samples, Normal populations; pooled t/ANOVA additionally assume equal population variances.

Which method fits the question?

Numerical outcome: one sample vs fixed reference → one-sample t; two independent groups → pooled t; three or more balanced groups → ANOVA. **Counts:** two event proportions → Yates z or 2×2 χ^2; larger table → uncorrected χ^2; one count list vs target probabilities → goodness of fit. **Follow-up:** requested pairs with correction. **Magnitude:** prospective risks → RR; case-control odds → OR.

State the population null, not sample equality

one-sample t	$H_0 : \mu = \mu_0$
pooled t	$H_0 : \mu_1 = \mu_2$
ANOVA	$H_0 : \mu_1 = \cdots = \mu_m$
two proportions	$H_0 : p_1 = p_2$

Contingency χ^2: same outcome distribution across groups (no association). **Goodness of fit:** each population category probability equals its specified target. **Two-sided alternatives:** the compared means/proportions differ; ANOVA means are not all equal.

Balanced ANOVA: supplied means and SDs

m (groups), n (subjects per group). **H₀:** population means equal. **Given:** $m = 3, n = 12$, means $1188, 1412, 1363$, SDs $189, 217, 191$. **1. Square SDs; average variances:**

$$s^2_{mathrmwit} = \frac{189^2 + 217^2 + 191^2}{3}$$

$$= \frac{35721 + 47089 + 36481}{3} = 39763.6667$$

2. Grand mean of the group means:

$$\bar{\bar{x}} = \frac{1188 + 1412 + 1363}{3} = 1321$$

3. Subtract grand mean; square and add:

$$(1188 - 1321)^2 + (1412 - 1321)^2 + (1363 - 1321)^2$$

$$= (-133)^2 + 91^2 + 42^2 = 27734$$

4. Variance of means; multiply by group size:

$$s^2_{\bar{x}} = \frac{27734}{3 - 1} = 13867$$

$$s^2_{mathrmbet} = n s^2_{\bar{x}} = 12(13867) = 166404$$

5. Divide; 6. df; 7. table; 8. conclude:

$$F = \frac{166404}{39763.6667} = 4.18483$$

$$\nu_1 = m - 1 = 2, \quad \nu_2 = m(n - 1) = 33$$

The supplied F table brackets the .05 critical value between 3.23 (df 40) and 3.32 (df 30). The statistic exceeds both: reject equal population means. Exact check: cutoff 3.28492, $p = .024001$. **Short route:** average variances; variance of means × n; divide. ANOVA alone does not identify the pair.

Holm: order, test, continue or stop

Order p smallest first. With matching distribution/df, order |t| or χ^2 largest first. Keep the pair labels attached.

$$\alpha_j = \frac{\alpha_T}{k - j + 1}$$

Reject → continue. First failure → stop all remaining. Never restart at a later, larger cutoff.

Pooled t: two independent means

s_p^2 (common estimated variance); SE_Δ (uncertainty of the mean difference).

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

$$\mathrm{SE}_\Delta = \sqrt{s_p^2(1/n_1 + 1/n_2)}$$

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\mathrm{SE}_\Delta}, \quad \nu = n_1 + n_2 - 2$$

Glucose: group 1 $(\bar{x}_1, s_1, n_1) = (82.2, 2.49, 25)$; group 2 $(86.1, 2.09, 25)$. **H₀:** $\mu_1 = \mu_2$.

$$s_1^2 = 6.2001, \quad s_2^2 = 4.3681$$

$$s_p^2 = \frac{24(6.2001) + 24(4.3681)}{48} = 5.2841$$

$$\mathrm{SE}_\Delta = \sqrt{5.2841(1/25 + 1/25)} = .650175$$

$$t = \frac{82.2 - 86.1}{.650175} = -5.99838$$

$$\nu = 48, \quad |t| \approx 6.00 > 2.011$$

Reject equal population means at .05. Control mean is lower. The sign describes direction; magnitude makes the two-sided decision. **Equal-size shortcut:** pooled variance is the average of the two variances.

Unequal sizes: keep the df weights

Cardiac index: group 1 $(2.08, 1.05, 9)$, group 2 $(1.75, .88, 16)$ in mean, SD, n order.

$$\nu = 9 + 16 - 2 = 23$$

$$s_p^2 = \frac{8(1.05^2) + 15(.88^2)}{23}$$

$$= \frac{8.82 + 11.616}{23} = .888522$$

$$\mathrm{SE}_\Delta = \sqrt{.888522(1/9 + 1/16)} = .392756$$

$$t = \frac{2.08 - 1.75}{.392756} = .840216$$

At df 23, two-tail .05 cutoff is 2.069; fail to reject. **Shortcut warning:** unequal sizes cannot use a simple unweighted average of variances.

Read real ANOVA output: nitric oxide

row	df	Mean Sq	F
Group	2	389.7	5.832
Residuals	19	66.8	

The printed Group-row p value is .0106. **1. Test:** overall population means. **2. Decision:** $.0106 < .05$, so reject equal means. **3. If asked to continue:** Residual Mean Sq supplies within variance 66.8; residual df is 19. **4. Pairwise family:** use requested groups and the stated correction; do not replace residual df with a new two-group df. The two mean-square entries are printed rounded, so their ratio need only agree approximately with the reported F.

Final arithmetic and conclusion checks

Units: mean and SD keep measurement units; variance squares them; test statistics and probabilities have no measurement units. **Margins:** expected counts must recover every observed row/column total. **Language:** reject supports a population difference; failing to reject does not prove equality. A p value is not the probability the null is true.

Degrees of freedom: quick method check

one-sample t	$n - 1$
pooled t	$n_1 + n_2 - 2$
ANOVA F	$(m - 1,\ m(n - 1))$
post-ANOVA pairs	$N - m$ (residual df)
table χ^2	$(r - 1)(c - 1)$
specified GOF	$k - 1$

One-sample t: fixed reference, not group 2

Heart rate: $\bar{x} = 76.8$, $s = 4.15$, $n = 5$; reference $\mu_0 = 72$ BPM. **H₀:** $\mu_{\mathrm{HR}} = 72$.

$$\mathrm{SEM} = \frac{s}{\sqrt{n}} = \frac{4.15}{\sqrt{5}} = 1.855936$$

$$\bar{x} - \mu_0 = 76.8 - 72 = 4.8$$

$$t = \frac{4.8}{1.855936} = 2.58630$$

$$\nu = n - 1 = 4, \quad t_{\mathrm{crit}} = 2.776$$

$2.58630 < 2.776$: fail to reject at .05; this small sample does not establish a population mean different from 72 BPM. **Short route:** difference from reference ÷ SEM. The fixed reference has no second sample variance.

After ANOVA: retain its shared noise and df

For requested groups i, j, use the ANOVA within variance:

$$|t_{ij}| = \frac{|\bar{x}_i - \bar{x}_j|}{\sqrt{s^2_{\mathrm{wit}}(1/n_i + 1/n_j)}}$$

$$\mathrm{SE}_{ij} = \sqrt{2s^2_{\mathrm{wit}}/n} \quad (\text{equal } n)$$

$$\nu = N - m = m(n - 1)$$

HDL groups: $n = 70$, $s^2_{\mathrm{wit}} = 239.9$, $\nu = 207$; means $64.8, 58.0, 43.3$.

$$\mathrm{SE}_{ij} = \sqrt{2(239.9)/70} = 2.61807$$

$$|t_{13}| = \frac{64.8 - 43.3}{2.61807} = 8.21216$$

$$|t_{23}| = \frac{58.0 - 43.3}{2.61807} = 5.61483$$

$$|t_{12}| = \frac{64.8 - 58.0}{2.61807} = 2.59733$$

All pairs: $k = m(m - 1)/2 = 3$. Against inactive control only: $k = m - 1 = 2$. Use the requested family, not automatically all pairs.

Bonferroni: one cutoff for the requested family

k (requested pairs), α_T (overall level):

$$\alpha_{\mathrm{pair}} = \frac{\alpha_T}{k}, \quad t_{\mathrm{crit}} = t_\nu^{-1}\left(1 - \frac{\alpha_T}{2k}\right)$$

Three HDL pairs: $\alpha_T = .05, k = 3, \nu = 207$.

$$\alpha_{\mathrm{pair}} = .016667, \quad t_{\mathrm{crit}} \approx 2.414$$

$$8.212 > 2.414, \quad 5.615 > 2.414, \quad 2.597 > 2.414$$

Reject all three pair nulls. Count only the requested comparisons. The two-tail t table uses the full individual level; inverse-t instead needs a left-area input.

Holm worked example: the stop rule matters

Given five comparisons with residual df 46, overall $\alpha_T = .05$. Sort the supplied |t| values with their pair labels:

j	pair	$\lvert t\rvert$	α_j	decision
1	1:6	3.3	.0100	reject
2	1:3	2.9	.0125	reject
3	1:5	2.3	.016667	stop
4	1:4	2.2	.0250	no reject
5	1:2	2.1	.0500	no reject

$$3.3 > 2.687, \qquad 2.9 > 2.600, \qquad 2.3 < 2.485$$

The third comparison fails, so the last three nulls are not rejected. Do not restart at the larger later thresholds. **Conclusion:** G1 differs from G6 and G3; differences from G5, G4 and G2 were not established.

Event proportions: label the numerator

m (events), n (total); code event=1, non-event=0.

$$\hat{p} = \frac{m}{n}, \quad \widehat{\mathrm{Var}}(X) = \hat{p}(1-\hat{p})$$

$$\mathrm{SE}(\hat{p}) = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

Use the same event in both groups. For death, count deaths; for thrombus, count thrombi. Binary variance and proportion SE are different: divide the former by n and square-root to obtain the latter.

Aspirin/placebo: corrected z, every step

Placebo: 18 thrombi/25; aspirin: 6/19. **H₀:** equal thrombus proportions. **1. Fractions; 2. pool actual counts:**

$$\hat{p}_1 = 18/25 = .72, \quad \hat{p}_2 = 6/19 = .315789$$

$$\hat{p} = \frac{18+6}{25+19} = \frac{24}{44} = .545455$$

3. Build the null SE:

$$\mathrm{SE}_0 = \sqrt{\hat{p}(1-\hat{p})(1/n_1 + 1/n_2)}$$

$$= \sqrt{.545455(.454545)(1/25+1/19)}$$

$$= \sqrt{.0229665} = .151547$$

4. Shrink the absolute gap by Yates:

$$c_Y = \tfrac{1}{2}(1/25 + 1/19) = .0463158$$

$$D_Y = \max(0, |.72 - .315789| - .0463158)$$

$$= .357895$$

5. Divide; 6. table and conclusion:

$$|z_Y| = .357895/.151547 = 2.36161$$

Infinity-df row of two-tail t table, alpha .05: 1.960. Reject equal event proportions: aspirin group has fewer thrombi. **Always use this correction for 2×2 tests.**

Same 2×2 table: expected cells and χ²

Rows placebo/aspirin; columns thrombus/no thrombus. Observed rows $(18, 7)$, $(6, 13)$; column totals $(24, 20)$, row totals $(25, 19)$, total 44.

$$E_{ij} = \frac{R_i C_j}{N}$$

	thrombus	no thrombus
placebo	$25(24)/44$	$25(20)/44$
aspirin	$19(24)/44$	$19(20)/44$

$$E = \begin{pmatrix} 13.63636 & 11.36364 \\ 10.36364 & 8.63636 \end{pmatrix}$$

Each absolute gap is 4.363636; corrected gap is 3.863636. Square once, divide by each expected count:

$$\chi^2_Y = \sum \frac{\max(0, |O-E| - .5)^2}{E}$$

$$= 1.094697 + 1.313636 + 1.440391 + 1.728469$$

$$= 5.577193, \quad \nu = (2-1)(2-1) = 1$$

At alpha .05 cutoff 3.841: reject. Arithmetic check: $2.36161^2 \approx 5.57719$. **Do not subtract .5 after squaring.**

Label a 2×2 table before taking ratios

	event	no event
exposed/treated	a	b
unexposed/control	c	d

$$r_T = \frac{a}{a+b}, \quad r_C = \frac{c}{c+d}, \quad \mathrm{RR} = \frac{r_T}{r_C}$$

$$\mathrm{OR} = \frac{a/b}{c/d} = \frac{ad}{bc}$$

Risk uses events/all; odds uses events/non-events. Reversing only one comparison takes the reciprocal. RR/OR of one means no association in that ratio.

Larger contingency table: no Yates

Observed depression rows: $(36, 131)$, $(41, 132)$, $(75, 104)$; row totals $167, 173, 179$, column totals $152, 367$, grand total 519. **H₀:** same outcome distribution across treatments.

$$E_{11} = 167(152)/519 = 48.90944$$

$$E = \begin{pmatrix} 48.90944 & 118.09056 \\ 50.66667 & 122.33333 \\ 52.42389 & 126.57611 \end{pmatrix}$$

$$\chi^2 = \sum_{i,j} \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

$$= 3.407393 + 1.411236 + 1.844298 + .763851$$

$$+9.722297 + 4.026673 = 21.175749$$

$$\nu = (3-1)(2-1) = 2$$

At .05, critical 5.991; reject the common-distribution null. Follow-ups use the requested 2×2 pairs, each with Yates, and a family correction. **Expected margins must match observed margins.**

Goodness of fit: one list vs target proportions

Exercise categories: observed $255, 125, 90$, total $N = 470$; target probabilities $.60, .25, .15$. **H₀:** category probabilities equal these targets.

$$E_i = Np_i \quad \Rightarrow \quad E = (282, 117.5, 70.5)$$

$$\chi^2 = \frac{(255-282)^2}{282} + \frac{(125-117.5)^2}{117.5}$$

$$+ \frac{(90-70.5)^2}{70.5}$$

$$= 2.585106 + .478723 + 5.393617 = 8.457447$$

$$\nu = k - 1 = 2, \quad 8.457447 > 5.991$$

Reject the target distribution at .05. **Expected = total × specified probability**; do not calculate expected cells from new row/column margins for this one-list question.

Supplied χ² output: compare only the requested pairs

Six pairs are printed, but only three control comparisons are requested. Each has df 1. Sort these χ² values strongest first: G3, G1, G2. Use Holm at overall .05:

$$\alpha_j = .05/3,\ .05/2,\ .05$$

$$18.234 > 6.635 > \chi^2_{.016667,1} > 5.024$$

$$9.4944 > \chi^2_{.025,1} = 5.024$$

$$4.6895 > \chi^2_{.05,1} = 3.841$$

Each passes in sequence: control differs from G3, G1 and G2. The first cutoff is bracketed using the supplied table columns. Do not add the unrequested pairs.

pnorm / qnorm: read inputs, predict the output

pnorm: measurement cutoff → cumulative area to its left. Inputs mean and sd describe the Normal distribution; sd is an SD, never a variance. **qnorm:** cumulative probability → measurement cutoff. **lower.tail=FALSE:** uses right-tail probability. No code writing is required.

For the glucose distribution (mean 95, SD 8.7):

$$\texttt{pnorm}(85, 95, 8.7) \approx .125190$$

$$\texttt{qnorm}(.975, 95, 8.7) \approx 112.0517$$

A mean-probability input needs the mean's SEM in place of individual SD. For supplied test output, read test name → statistic → df → p → contextual decision.

Quantify size: RR, ARR, NNT, OR

For aspirin/placebo thrombi, treatment risk $r_T = 6/19 = .315789$, control risk $r_C = 18/25 = .72$.

$$\mathrm{RR} = \frac{r_T}{r_C} = \frac{6/19}{18/25} = .438596$$

Aspirin's thrombus risk is about .44 times placebo risk; it is not a 44-percentage-point reduction.

$$\mathrm{ARR} = r_C - r_T = .72 - .315789 = .404211$$

$$\mathrm{NNT} = 1/\mathrm{ARR} = 2.47396$$

About 40.42 percentage points lower absolute risk; whole-person NNT rounds up to 3. For adverse-event increase, NNH is one divided by that increase. **Prospective trial/cohort:** group risks are estimable. **Case-control:** cases selected by outcome; use odds, not sample disease risk. Case-control counts: cases exposed/unexposed 50/14; controls 43/35.

$$\mathrm{OR} = \frac{50/14}{43/35} = \frac{50(35)}{14(43)} = 2.90698$$

Exposure odds are about 2.91 times as high among cases as controls. Risk uses events/all; odds uses events/non-events.

Design and quick interpretation checks

Randomized clinical trial: assigns treatments, follows outcomes (prospective). **Prospective cohort:** no randomization; follow exposure groups. **Retrospective case-control:** select people with/without outcome, look back at exposure. Randomization concerns assignment; masking concerns who knows it. **Association is not automatically causation.** Ask separately: is a difference established (test), and how large is it (RR/OR/absolute change)?

Binary SE: keep the two spreads separate

Halothane example: 8 deaths among 61 patients. The event is death. **1. Estimate proportion; 2. binary variance:**

$$\hat{p} = \frac{8}{61} = .131148$$

$$\hat{p}(1-\hat{p}) = \frac{8}{61}\frac{53}{61} = .113948$$

3. Individual binary SD; 4. SE of the proportion:

$$\widehat{\mathrm{SD}}(X) = \sqrt{.113948} = .337562$$

$$\mathrm{SE}(\hat{p}) = \sqrt{.113948/61} = .0432203$$

A larger n reduces uncertainty in the proportion. If everyone has the event or nobody does, the binary observations have no spread.

Relative reduction can hide a small absolute benefit

Death risk drops from 10 in 1000 to 8 in 1000:

$$r_C = .010, \quad r_T = .008, \quad \mathrm{RR} = .8$$

$$\mathrm{RRR} = 1 - \mathrm{RR} = .20 \quad (20\% \text{ relative reduction})$$

$$\mathrm{ARR} = .010 - .008 = .002, \quad \mathrm{NNT} = \frac{1}{.002} = 500$$

The absolute reduction is .2 percentage points, not 20 points. If serious-harm NNH is 20, then:

$$500/20 = 25 \text{ harms per additional death avoided}$$

For an adverse-event increase, $r_T > r_C$:

$$\mathrm{NNH} = \frac{1}{r_T - r_C}$$

Use probabilities in reciprocals. NNT/NNH is tied to the named outcome and follow-up; round upward if a whole-person count is requested.